Grade 11 · Algebra II · Topic 6: Exponential and Logarithmic Functions · Lesson 6-6 — Exponential and Logarithmic Equations and Inequalities

Project Task — Red Sea Resort Planning

A 20-minute group project. Work in fours. Every member signs the sheet.

Student name :	Date : .....
Group members :	

MATERIALS This sheet · Squared paper · A pencil and ruler · A calculator

THE DRIVING QUESTION

Two model resorts grow in different ways, and water use per visitor falls each year. When does each reach its target, and what should the planners do first?

THE SITUATION

Vision 2030 includes new Red Sea tourism resorts. These are model values for practice, not forecasts.

Here t is the number of years after opening. Each resort has a capacity of 300 000 visitors per year, and planners want water use per visitor below 50 litres.

$$A(t) = 40\,000\,(1.25)^t \quad \text{visitors per year}$$

$$B(t) = 20\,000 \cdot 2^{t/3} \quad \text{visitors per year}$$

$$W(t) = 120\,(0.8)^t \quad \text{litres per visitor}$$

Years after opening	0	3	6	9
Resort A visitors				
Resort B visitors				

YOUR TASK 5 steps · 20 minutes

1 FILL THE TABLE (4 min)
Complete both rows of the table using the models, and show one calculation for each resort.

2 NAME THE METHOD (4 min)
For Resort A to reach 100 000 visitors, write the exact answer as a logarithm and then give it to 1 decimal place. For Resort B to reach 80 000, say why a common base works and find the time.

3 CAPACITY (5 min)
For how many years is each resort below 300 000 visitors per year? Write the inequality, keep its direction, and give each answer to 1 decimal place. Which resort needs Phase 2 first?

4 THE WATER TARGET (4 min)
Find when water use per visitor first falls below 50 litres. The base is below 1: say what happens to the inequality, and give your answer in whole years.

5 PRESENT (3 min)
One sentence for the planners: which resort first, when, and how you know your answers are valid.

YOUR WORKING write on this sheet

Step 1 — the table, one calculation each:

Step 2 — exact logarithm, then decimal:

Step 3 — the two capacity inequalities:

Step 4 — the water inequality and the flip:

Step 5 — our sentence:

ROLES

- Reader — states the models and the targets
- Calculator — fills the table and the decimals
- Checker — tests each answer in the original equation or inequality
- Presenter — says the sentence

DONE WHEN ...

- Every answer is checked by substituting it back into the model.
- The exact logarithm is written before the decimal.
- The direction of each inequality is justified by the size of the base.
- Your sentence makes sense to someone who does not study maths.